

KDR-CA-AEAD User Manual & Operational Reference

Official End-User Manual & Reproducibility Reference — Version 1.0.0

1. Executive Summary & Version Metadata

Project: KDR-CA-AEAD v1.0.0 | Python 3.13.5 | Windows/Linux/macOS | Main Branch

The KDR-CA-AEAD framework delivers authenticated encryption with associated data combining Keyed Dynamically-Reconfigured Cellular Automata permutations with HKDF-SHA256 and HMAC-SHA256.

2. Quick Execution Commands

- Run Pytest Suite: `python -m pytest`
- Build Figures: `python scripts/generate_architecture_figures.py`
- Build Graphs: `python scripts/generate_benchmark_graphs.py`
- Build Paper: `python paper/build_paper.py`